

Dae Seok Song

AI Research Engineer | Brain-inspired and Modular Learning
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Seoul, Republic of Korea

Research Profile

AI research engineer who has led model development, optimization, clinical-product integration, and international regulatory evidence generation at a medical-AI startup since November 2022. My research asks how a trained neural system can acquire, exchange, and preserve functional modules without global retraining. I combine reproducible experiments with deployment constraints learned from multi-institution medical imaging.

Research Interests

Brain-inspired learning; modular and continual learning; representation alignment; neural population geometry; robust learning under distribution shift.

Independent Research

RevolverNet: Detachable Functional Modules and Conditional Routing

2026–Present

Independent Researcher

Seoul, Korea

- Designed a modular architecture with interchangeable slots, detachable functional modules, and a router; the project metaphor names these chambers, bullets, and a trigger.
- Across two completed image-classification conditions, one-time representation alignment recovered 99.4–99.8% of the performance lost under naive swaps, covering every directed donor-to-recipient pair among four separately trained models and passing the pre-specified acceptance criterion in every transfer tested. The models shared the dataset, architecture, and role assignment but differed in initialization. Preliminary, unpublished independent work.
- Implemented hard routing so the trigger selects one bullet for each input. Planned work will test whether inputs not covered by existing bullets can indicate when a new chamber and bullet are needed.
- Long-term goal: let people inspect, remove, and replace functional modules while preserving validated behavior and avoiding unnecessary computation.

Professional Experience

Connecte Co., Ltd.

Nov 2022–Present

AI Research Engineer / Early Team Member

Seoul, Korea

- Consolidated 10 production knee X-ray models into 2 uncertainty-aware, partial-label multi-task models: stored weights 5.56 GB $\rightarrow$ 0.51 GB (10.97 $\times$) and inference latency 2.5 $\times$ lower, with F-score kept within 0.003 of the original models.
- Rebuilt the optimization-to-serving line (ONNX optimization, 16-bit inference, BentoML serving): engine size 47% smaller and inference 3 $\times$ faster, with clinical validation showing only a 0.06-percentage-point accuracy change and 2.1% grade changes (non-inferiority confirmed by patient-level bootstrap).
- Tuned per-grade decision thresholds with Youden's J statistic across four tie-break strategies, surpassing an FDA-cleared competitor product's F-score on a matched evaluation.
- Built the standard PyTorch Lightning + Hydra experiment pipeline for medical foundation models (DINOv3, Rad-DINO) with selectable linear probing or LoRA, over 100,000+ preprocessed and indexed medical records.
- Built an in-house MLOps console (GUI/CLI) with a two-stage hyperparameter search (LLM-agent proposals, then Optuna/TPE on plateau) that raised detection mAP by 6.9 points (0.80 $\rightarrow$ 0.87); evaluated 12 agentic-workflow frameworks and selected LangGraph, running agents on vLLM (Qwen, Mistral); caching cut re-onboarding from 3 h 36 min to 38 min (5.7 $\times$).
- Led AI technical documentation for medical-device approval in Korea (MFDS), the EU (CE MDR), the UAE (DOH), and Singapore (HSA).

Minerva Soft

Mar 2022–Jul 2022

Image Processing Intern

Korea

- Replaced full-image barcode scanning with an OpenCV ROI pipeline, raising recognition from 70% to 90%; shipped a C++ DLL after 170,000+ memory-leak and stability iterations.

Education

University of Suwon
B.S. in Information Security, Valedictorian

Mar 2017–Feb 2023
GPA 4.29/4.50; Major 4.37/4.50

Selected Publication

Song, D. S. et al. “Accurate, automated classification of radiographic knee osteoarthritis severity using a novel method of deep learning: Plug-in modules.” *Knee Surgery & Related Research*, 2024. Second author; devised and implemented the core plug-in-module ensemble that imitates clinicians’ consensus, with the paper reporting an accuracy improvement of about 6%. [doi:10.1186/s43019-024-00228-3](https://doi.org/10.1186/s43019-024-00228-3)

Patents & Other Research Output

- Inventor, Korean Patent Application 10-2024-0124529 and PCT/KR2025/012689: ensemble system for knee-osteoarthritis severity classification (the plug-in-module method above).
- Inventor, Korean Patent Application 10-2025-0043898: generating a patient’s projected knee image from the current radiograph and its K–L/OARSI grades.
- First-author study submitted to KSC 2022: geometric preprocessing alone raised lesion-segmentation DICE from 86.95 to 90.48.

Technical Strengths

Research	Controlled ablation design, multi-seed evaluation, representation analysis, patient-level bootstrap non-inferiority tests, reproducibility
ML	PyTorch, PyTorch Lightning, Hydra, multi-task and partial-label learning, foundation-model adaptation (linear probing, LoRA), medical-image classification/segmentation
Compression & serving	ONNX, OpenVINO 16-bit, quantization and pruning, TorchServe, BentoML, FastAPI, vLLM
Engineering	Python, C/C++, Docker, GPU server operation (H100, L40S), regulated-product documentation

Languages

Korean (native); English (conversational); Japanese (conversational).